Pillar 1 · advantage estimator

VARY — the intended advantage-estimator contrast. The executed PPO/GRPO comparison also differs in prompt exposure and evaluation overlap (Chapter 6).

FIX — model checkpoint, dataset slice, decoding path, reward grader, framework, schedule and seed set.

Measured outcome. *Paired* $\Delta = -0.002$, $p = 0.374$*: the observed contrast is small and not significant; residual design differences limit attribution.*

Pillar 2 · signal starvation

VARY — the internal signal state of the batch: per-step Zero-Variance Fraction and its collapse decomposition, logged at every step rather than summarised after the run.

FIX — the training stack is untouched; the diagnostic observes, it does not intervene.

Measured outcome. 0.16 *mean ZVF on Qwen3-8B / GSM8K at* $G = 8$*. All nine rows of the method panel, vanilla GRPO included, are a declared simulation projection.*

Pillar 3 · group size

VARY — the group size $G \in \{2, 4, 8, 16\}$ across seeds, holding model, data, reward and schedule fixed.

FIX — model, data, reward grader, learning-rate schedule and evaluation split.

Measured outcome. *Mean ZVF falls* $0.838 \rightarrow 0.631$ *(*$G = 2 \rightarrow 16$*); the training-reward ratio to* $G = 2$ *is* ≈ 1.00*, equivalent up to* $G = 8$ *but not established at* $G = 16$*.*

Pillar 4 · held-out behaviour

VARY — the method, Dr. GRPO versus GRPO, compared pre-RL against post-RL on data the training loop never saw.

FIX — the pre-RL checkpoint and the held-out split, evaluated side by side under frozen sampling.

Measured outcome. *Between-method held-out gain difference is not significant; both pre/post gains are positive. Length drifts* $194.4 \rightarrow 183.5$ *tokens under the cap.*

Intended controls; executed departures accompany each result.

Study-specific settings (Table 4.2). There is no single common configuration or test for all four pillars. The arithmetic PPO/GRPO comparison changes exposure and evaluation overlap; the group-size sweep fixes prompts and steps rather than total sampled responses. Seed counts, loss implementations, evaluation populations and uncertainty methods are reported with each result.